Group-9

4J 7H 9P FU 2h CE 3d

			Display 1								Display 2							
x	y	z	D/L	a	b	c	d	e	f	g	D/L	a_0	b_0	c_0	d_0	e_0	f_0	g_0
0	0	0	4	0	1	1	0	0	1	1	J	0	1	1	1	1	0	0
0	0	1	7	1	1	1	0	0	0	0	H	0	1	1	0	1	1	1
0	1	0	9	1	1	1	1	0	1	1	P	1	1	0	0	1	1	1
0	1	1	F	1	0	0	0	1	1	1	U	0	1	1	1	1	1	0
1	0	0	2	1	1	0	1	1	0	1	h	0	0	1	0	1	1	1
1	0	1	C	1	0	0	1	1	1	0	E	1	0	0	1	1	1	1
1	1	0	3	1	1	1	1	0	0	1	d	0	1	1	1	1	0	1
1	1	1		x	x	x	x	x	x	x		x	x	x	x	x	x	x

Generalized Equations & Simplified Equations:

$a = \Sigma(1,2,3,4,5,6)$

$= x'y'z + x'yz' + x'yz + xy'z' + xy'z + xyz'$

Generalized (SOP)

$= \Pi(0)$

$= x+y+z$ (POS) (Generalized)

K-map:

$x'y'z' = (x+y+z)$

	$y'z'$	$y'z$	yz	yz'
x'	0_0	1_1	1_3	1_2
x	1_4	1_5	X_7	1_6

x z y

$a = x + y + z$ (Simplified SOP)

$= x + y + z$ (Simplified POS)

$b = \Sigma(0,1,2,4,6)$

$= x'y'z' + x'y'z + x'yz' + xy'z' + xyz'$ (Generalized SOP)

$= \Pi(3,5)$

$= (x+y'+z')(x'+y+z')$ (Generalized POS)

K-map:

$x'y'$ z'

	$y'z'$	$y'z$	yz	yz'
x'	1_0	1_1	0_3	1_2
x	1_4	0_5	X_7	1_6

$yz \rightarrow y'+z'$

$xz \rightarrow x'+z'$

$b = x'y' + z'$ (Simplified SOP)

$= (x'+z')(y'+z')$ (Simplified POS)

$C = \Sigma(0,1,2,6)$

$= x'y'z' + x'y'z + x'yz' + xyz'$ (Generalized SOP)

$= \Pi(3,4,5)$

$= (x+y'+z')(x'+y+z)(x'+y+z')$

K-map:

	$y'z'$	$y'z$	yz	yz'
x'	1_0	1_1	0_3	1_2
x	0_4	0_5	X_7	1_6

$\to x'y'$

$\to yz'$

$yz \to y'+z'$

$xy' \to (x'+y)$

$C = x'y' + yz'$ (Simplified SOP)

$= (x'+y)(y'+z')$ (Simplified POS)

$d = \Sigma(2,4,5,6)$

$= x'yz' + xy'z' + xy'z + xyz'$

(Generalized SOP)

$= \Pi(0,1,3)$

$= (x+y+z)(x+y+z')(x+y'+z')$

(Generalized POS)

k-map:

$x'y' \to (x+y)$

$x'z \to x+z'$

$\to yz'$

	$y'z'$	$y'z$	yz	yz'
x'	0_0	0_1	0_3	1_2
x	1_4	1_5	X_7	1_6

$\to x$

$d = x + yz'$ (Simplified SOP)

$= (x+y)(x+z')$ (Simplified POS)

$e = \Sigma(3,4,5) = x'yz + xy'z' + xy'z$

(Generalized SOP)

$\cancel{= \Pi(}$

$= \Pi(0,1,2,6)$

$= (x+y+z)(x+y+z')(x+y'+z)(x'+y'+z)$

(Generalized POS)

k-map:

$x'y' \to (x+y)$

	$y'z'$	$y'z$	yz	yz'
x'	0_0	0_1	1_3	0_2
x	1_4	1_5	X_7	0_6

$\to yz$

$\to yz' \to y'+z$

$\to xy'$

$e = xy' + yz$ (Simplified SOP)

$= (x+y)(y'+z)$ (Simplified POS)

$f = \Sigma(0, 2, 3, 5)$

$= x'y'z' + x'yz' + x'yz + xy'z$ (Generalized ~~POS~~ SOP)

$= \Pi(1, 4, 6)$

$= (x+y+z')(x'+y+z)(x'+y'+z)$ (Generalized POS)

k-map:

$\to x'y'z \to (x+y+z')$

	$y'z'$	$y'z$	yz	yz'
x'	1_0	0_1	1_3	1_2
x	0_4	1_5	X_7	0_6

$\to x'z'$

$\to x'y$

$\to xz$

$\to xz' \to (x'+z)$

$f = x'z' + x'y + xz$ (Simplified SOP)

$= (x+y+z')(x'+z)$ (Simplified POS)

$g = \Sigma(0,2,3,4,6)$

$= x'y'z' + x'yz' + x'yz + xy'z' + xyz'$ (Generalized ~~POS~~ SOP)

$= \Pi(1,5)$

$= (x+y+z')(x'+y+z')$ (Generalized ~~SOP~~ POS)

K-map:

	$y'z'$	$y'z$	yz	yz'
x'	1_0	0_1	1_3	1_2
x	1_4	0_5	X_7	1_6

$\to y$

$\to z'$

$y'z \to (y+z')$

$g = y + z'$ (Simplified SOP)

$= (y+z')$ (Simplified POS)

$a_0 = \Sigma(2,5)$

$= x'yz' + xy'z$ (Generalized SOP)

$= \Pi(0,1,3,4,6)$

$= (x+y+z)(x+y+z')(x+y'+z')(x'+y+z)$

$(x'+y'+z)$ (Generalized POS)

	$y'z'$	$y'z$	yz	yz'
x'	0_0	0_1	0_3	1_2
x	0_4	1_5	X_7	0_6

$\to x'z \to (x+z')$

$\to x'yz'$

$\to xz' \to (x'+z)$

$\to xz$

$\to y'z' \to (y+z)$

$a_0 = xz + x'yz'$ (Simplified SOP)

$= (x'+z)(x+z')(y+z)$ (Simplified POS)

$b_0 = \Sigma(0,1,2,3,6)$

$= x'y'z' + x'y'z + x'yz' + x'yz + xyz'$

(Generalized SOP)

$= \Pi(4,5)$

$= (x'+y+z)(x'+y+z')$ (Generalized POS)

k-map:

	$y'z'$	$y'z$	yz	yz'
x'	1_0	1_1	1_3	1_2
x	0_4	0_5	X_7	1_6

$\to x'$

$\to y$

$\to xy' \to (x'+y)$

$b_0 = x'+y$ (Simplified SOP)

$= (x'+y)$ (Simplified POS)

$C_o = \Sigma(0, 1, 3, 4, 6)$

$= x'y'z' + x'y'z + x'yz + xy'z' + xyz'$

(Generalized SOP)

$= \Pi(2, 5)$

$= (x + y' + z)(x' + y + z')$ (Generalized POS)

K-map:

	$y'z'$	$y'z$	yz	yz'
x'	1 ₀	1 ₁	1 ₃	0 ₂
x	1 ₄	0 ₅	X ₇	1 ₆

$\to x'yz' \to (x+y'+z)$

$\to x'z$

$\to xz'$

$\to y'z'$

$\to xz \to (x'+z')$

$C_o = x'z + y'z' + xz'$ (Simplified SOP)

$= (x + y' + z)(x' + z')$ (Simplified POS)

$d_o = \Sigma(0, 3, 5, 6)$

$= x'y'z' + x'yz + xy'z + xyz'$

(Generalized SOP)

$= \Pi(1, 2, 4)$

$= (x+y+z')(x+y'+z)(x'+y+z)$

(Generalized POS)

	$y'z'$	$y'z$	yz	yz'
x'	1_0	0_1	1_3	0_2
x	0_4	1_5	$\cancel{X}_7$	1_6

$x'y'z$

$\rightarrow x'yz' \rightarrow (x+y'+z)$

$\rightarrow yz$

$\rightarrow xy$

$(x'+y+z) \leftarrow xy'z'$ $\quad x'y'z \rightarrow (x+y+z')$ $\quad \rightarrow xz$

$$d_0 = x'y'z' + xy + xz + yz \quad \text{(Simplified SOP)}$$

$$= (x+y'+z)(x+y+z')(x'+y+z) \quad \text{(Simplified POS)}$$

$$e_0 = \Sigma(0,1,2,3,4,5,6)$$

$$= x'y'z' + x'y'z + x'yz' + x'yz + xy'z' + xy'z + xyz' \quad \text{(Generalized SOP)}$$

$$= 1 \quad \text{(Generalized POS)}$$

K-map:

	$y'z'$	$y'z$	yz	yz'
x'	1_0	1_1	1_3	1_2
x	1_4	1_5	X_7	1_6

$\rightarrow 1$

$$e_0 = 1 \quad \text{(Simplified SOP)}$$

$$= 1 \quad \text{(Simplified POS)}$$

$f_0 = \Sigma(1,2,3,4,5)$

$= x'y'z + x'yz' + x'yz + xy'z' + xy'z$

(Generalized SOP)

$= \cancel{\Pi(}$

$= \Pi(0,6)$

$= (x+y+z)(x'+y'+z)$ (Generalized POS)

k-map:

$\to x'y'z' \to (x+y+z)$

	$y'z'$	$y'z$	yz	yz'
x'	0_0	1_1	1_3	1_2
$\cancel{x}$	1_4	1_5	X_7	0

$\to x'y$

$\to xy$

$\hookrightarrow x'+y'$

$\hookrightarrow z$

$\downarrow xy'$

(Simplified SOP)

$f_0 = x'y + xy' + z$

$= (x+y+z)(x'+y')$ (Simplified POS)

$g_0 = \Sigma(1,2,4,5,6)$

$= x'y'z + x'yz' + xy'z' + xy'z + xyz'$)

(Generalized SOP)

$= \Pi(0,3)$

$= (x+y+z)(x+y'+z')$ (Generalized POS)

$x'y'z' \rightarrow (x+y+z)$

	$y'z'$	$y'z$	yz	yz'
x'	0_{0}	1_{1}	0_{3}	1_{2}
x	1_{4}	1_{5}	X_{7}	1_{6}

$\rightarrow yz'$

$\rightarrow y'z$

$\rightarrow x$

$\rightarrow yz \rightarrow y'+z'$

$g_0 = x + y'z + yz'$ (Simplified SOP)

$= (x+y+z)(y'+z')$ (Simplified POS)